

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 9

Question 1 - MCQ · PG-42 · LR-1

A number of the form $a + ib$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$, is called

Options

- A. a complex number
- B. an irrational number
- C. an imaginary number only
- D. a natural number

Correct answer: a complex number

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives a complex number.

Question 2 - MCQ · PG-42 · LR-2

If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1 + z_2$ equals

Options

- A. $(a + c) + (b + d)i$
- B. $(ac - bd) + (ad + bc)i$
- C. $(a - c) + (b - d)i$
- D. $(a + b) + (c + d)i$

Correct answer: $(a + c) + (b + d)i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(a + c) + (b + d)i$.

Question 3 - MCQ · PG-42 · LR-3

If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1 z_2$ equals

Options

- A. $(ac - bd) + (ad + bc)i$
- B. $(ac + bd) + (ad - bc)i$
- C. $(a + c) + (b + d)i$
- D. $(ac - bd) + (ab + cd)i$

Correct answer: $(ac - bd) + (ad + bc)i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(ac - bd) + (ad + bc)i$.

Question 4 - MCQ · PG-42 · LR-4

For a positive integer k , i^{4k} equals

Options

- A. 1
- B. -1
- C. i
- D. $-i$

Correct answer: 1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 1.

Question 5 - MCQ · PG-42 · LR-5

For a positive integer k , i^{4k+1} equals

Options

- A. i
- B. 1
- C. -1
- D. $-i$

Correct answer: i

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives i .

Question 6 - MCQ · PG-42 · LR-6

For a positive integer k , i^{4k+2} equals

Options

- A. -1
- B. 1
- C. i
- D. $-i$

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1.

Question 7 - MCQ · PG-42 · LR-7

For a positive integer k , i^{4k+3} equals

Options

- A. $-i$
- B. i
- C. 1
- D. -1

Correct answer: $-i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $-i$.

Question 8 - MCQ · PG-42 · LR-8

The conjugate of $z = a + ib$ is

Options

- A. $\bar{z} = a - ib$
- B. $\bar{z} = -a + ib$
- C. $\bar{z} = a + ib$
- D. $\bar{z} = b + ia$

Correct answer: $\bar{z} = a - ib$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\bar{z} = a - ib$.

Question 9 - MCQ · PG-42 · LR-9

The cube roots of unity are denoted by

Options

- A. $1, w, w^2$ or $1, w, \bar{w}$
- B. $1, i, -i$
- C. $0, w, w^2$
- D. $1, w^2, w^3$

Correct answer: $1, w, w^2$ or $1, w, \bar{w}$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $1, w, w^2$ or $1, w, \bar{w}$.